

Quadratic Number Theory

Tanvir Ahmed

1 Introduction

While “quadratic” number theory isn’t a conventional name for a particular subject in number theory, it’s helpful to think of number theory that is related to quadratics/squares as related to each other. In this article, we will look at the notion of quadratic residues and devise different ways to check if something is a quadratic residue, and then apply this knowledge to the square-sum problem, which in general concerns representing positive integers as a sum of perfect squares. In the first section, we will introduce the notion of a field as it is a helpful way to think about congruences and modular arithmetic.

2 The Finite Fields of Order p

2.1 Definition of a Field

It is first important to establish that the set of numbers from 0 to $p - 1$ for a prime number p forms a field under addition and multiplication mod p . We first have to define a field, which is as follows:

Definition 2.1. A field F is a set of elements, along with two binary operations $+$ and $\cdot$ (addition and multiplication) such that the following are true for all $a, b, c \in F$:

- $a + b = b + a$
- $a \cdot b = b \cdot a$
- $a + (b + c) = (a + b) + c$
- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
- $a \cdot (b + c) = a \cdot b + a \cdot c$
- $a + 0 = a$ for some fixed element $0 \in F$ (called the additive identity)
- $a \cdot 1 = a$ for some fixed element $1 \in F$ (called the multiplicative identity)
- There is an element $-a \in F$ such that $a + (-a) = 0$
- If $a \neq 0$, there is an element $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$

Some common examples of fields are $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{C}$, under the usual operations of addition and multiplication. Now, we let $\mathbb{Z}_n$ be the set of numbers from 0 to $n - 1$ inclusive, where the operations of addition and multiplication are defined mod n . For example, in $\mathbb{Z}_6$, $3 + 4 = 1$, because once you reduce $3 + 4 \bmod 6$, you get 1, and multiplication is defined similarly. For the rest of this paper, we will often call the elements of $\mathbb{Z}_n$ residues. It is easy to check that all the properties of a field except for the last one hold for all positive integers n for $\mathbb{Z}_n$. Considering when the last property is true leads to the following theorem.

Theorem 2.1. $\mathbb{Z}_n$ is a field if and only if n is prime.

Proof. The reverse direction is easy to prove. Suppose n is composite, and let a be a divisor of n not equal to 1 or n . Then, for all integers b , ab will be a multiple of a , a property that is preserved mod n because n is also a multiple of a . In other words, ab cannot be equal to 1 for any b , so a can't have an inverse mod n .

Suppose n is prime and let a be an integer such that $a \not\equiv 0 \pmod{n}$. We wish to show that a has an inverse mod n . The condition that $a \not\equiv 0 \pmod{n}$ is equivalent to saying that a is relatively prime to n since n is prime. Thus, by Bezout's Lemma, there are integers b and k such that $ab + nk = 1$. Reducing this mod n , we get $ab \equiv 1 \pmod{n}$, which shows that b is the inverse of a mod n . $\square$

This implies that $\mathbb{Z}_p$ is a field for prime p . These fields are known as finite fields, as opposed to $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{C}$, etc. which are infinite. The second part of this proof also implies that in general, if $\gcd(a, n) = 1$, a has a multiplicative inverse mod n , which is also important.

2.2 Polynomials

Definition 2.2. A polynomial $p(x)$ in the field F is of the form $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ where $a_i \in F$, $a_n \neq 0$, and x is a variable. Furthermore, the degree of $p(x)$ is n . If $a \in F$, then $p(a)$ is the result of substituting a for x in the expression $p(x)$ and evaluating it. The set of polynomials over F is denoted by $F[x]$, and is called the ring of polynomials over F .

Most of the time, when we deal with polynomials, we are most concerned with solving for the values of x which make the polynomial evaluate to 0, also called the roots of the polynomial. For real numbers, it is well known that if a polynomial has degree n , it has at most n roots, and this is true for fields of the form $\mathbb{Z}_p$ as well, which we will now prove.

Theorem 2.2. If $p(x) \in F[x]$ for an arbitrary field F has degree n , $p(x)$ has at most n roots.

Proof. We prove this by induction on n . If $n = 0$, $p(x)$ is always equal to a nonzero constant, so it has no solutions. If $n = 1$, $p(x) = ax + b$ for $a, b \in F$. In this case, $ax + b = 0$ implies $x = -\frac{b}{a}$, so $p(x)$ only has one solution.

Now suppose that all polynomials of degree n have at most n roots. Let $p(x)$ be a polynomial of degree $n + 1$. We can express it as $p(x) = a_{n+1} x^{n+1} + a_n x^n + \cdots + a_1 x + a_0$. If $p(x)$ has no roots in F , we are done. otherwise, suppose x_1 is a root. Then, we have the following two equations for all $x \in F$:

$$\begin{aligned} p(x) &= a_{n+1} x^{n+1} + a_n x^n + \cdots + a_1 x + a_0 \\ 0 &= a_{n+1} x_1^{n+1} + a_n x_1^n + \cdots + a_1 x_1 + a_0 \end{aligned}$$

Subtracting these equations gives us:

$$p(x) = a_{n+1}(x^{n+1} - x_1^{n+1}) + a_n(x^n - x_1^n) + \cdots + a_1(x - x_1)$$

Importantly, for each i , we have $x^i - x_1^i = (x - x_1)(x^{i-1} + x^{i-2}x_1 + \cdots + x_1^{i-1})$. We will call that second factor $q_i(x)$ because it's not important what it is. Therefore, $x^i - x_1^i = (x - x_1)q_i(x)$. Plugging this back into the last equation, we get

$$\begin{aligned} p(x) &= a_{n+1}(x - x_1)q_{n+1}(x) + a_n(x - x_1)q_n(x) + \cdots + a_1(x - x_1)q_1(x) \\ &= (x - x_1)(a_{n+1}q_{n+1}(x) + \cdots + a_1q_1(x)) \end{aligned}$$

If we call the second factor $q(x)$, we get $p(x) = (x - x_1)q(x)$, where $q(x)$ has degree n . Thus, if x is a root of $p(x)$, x is a root of $x - x_1$ or of $q(x)$. $x - x_1$ has only one root, and $q(x)$ has at most n roots by the induction hypothesis, so altogether, $p(x)$ has at most $n + 1$ roots. □

3 Quadratic Residues

3.1 QR's and QNR's

The properties of a field are such that one can easily solve linear equations (in the context of $\mathbb{Z}_p$, they would be linear congruences), and we do this by using the standard four operations (addition, subtraction, multiplication, division). Quadratic equations, however, are not as easy, and it is not guaranteed that they always have a solution. In the next few sections, we are interested in when quadratic equations have solutions in the field $\mathbb{Z}_p$. Since equations over $\mathbb{Z}_p$ are actually congruences, we will refer to these as quadratic congruences.

The simplest quadratic congruence is $x^2 \equiv a \pmod{p}$ for some $a \in \mathbb{Z}_p$. We say that a is a quadratic residue mod p if $a \not\equiv 0 \pmod{p}$ and this congruence has a solution. Since $\mathbb{Z}_p$ is a field and this congruence is a polynomial of degree 2, the congruence has at most 2 solutions in x . In fact, if $a \not\equiv 0 \pmod{p}$, and the congruence has one solution, then it has 2 solutions, because if $x^2 \equiv a \pmod{p}$, $(-x)^2 \equiv a \pmod{p}$ as well. We will use the abbreviations QR and QNR to stand for quadratic residue and quadratic non-residue respectively.

The first important thing we can establish about quadratic residues is that there is always an equal numbers of QR's and QNR's for any odd prime p .

Proposition 3.1. If p is an odd prime, there are exactly $\frac{p-1}{2}$ QR's and $\frac{p-1}{2}$ QNR's.

Proof. Every quadratic residue corresponds to two "square roots", and these square roots must be distinct. To see this, suppose that the congruence $x^2 \equiv a \pmod{p}$ has two identical roots for some QR a and say that root is r . We saw how these roots are negatives of each other, which implies that $r \equiv -r \pmod{p} \implies 2r \equiv 0 \pmod{p}$. Since a is a QR, $a \not\equiv 0 \pmod{p}$, so $r \not\equiv 0 \pmod{p}$. By reducing $r \pmod{p}$, we may then assume $1 \leq r \leq p-1 \implies 2 \leq 2r \leq 2p-2$. By the congruence $2r \equiv 0 \pmod{p}$, we have that $2r$ is a multiple of p , but the only multiple of p in the range 2 to $p-2$ is p , so $2r = p$, but p is odd, leading to a contradiction.

We also know that every residue from 1 to $p-1$ is a square root of some QR: it is a square root of its square. Therefore, if we consider all the square roots of all QR's, not only do we get all residues from 1 to $p-1$, but we also get twice the number of QR's, implying that the number of QR's is $\frac{p-1}{2}$, and the number of QNR's is $p-1 - \frac{p-1}{2} = \frac{p-1}{2}$. □

3.2 The Legendre Symbol

The Legendre symbol, denoted by $\left(\frac{a}{p}\right)$ is defined for odd primes p as follows:

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & a \text{ is a QR mod } p \\ -1 & a \text{ is a QNR mod } p \\ 0 & a \equiv 0 \pmod{p} \end{cases}$$

One important property of the Legendre Symbol is that it is multiplicative, which we will now prove.

Proposition 3.2. For an odd prime p and for integers a and b , $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$.

Proof. If $a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$, then $\left(\frac{ab}{p}\right) = \left(\frac{0}{p}\right) = 0$ and $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = 0$, so the two expressions are equal.

If $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right) = 1$, then $a \equiv x^2 \pmod{p}$, $b \equiv y^2 \pmod{p}$, so $ab \equiv (xy)^2 \pmod{p}$, so $\left(\frac{ab}{p}\right) = 1$, making the two expressions equal.

If one of a and b is a QR and the other is a QNR, then without loss of generality, say $a \equiv x^2 \pmod{p}$ and b is QNR. If ab is a QR, so that $ab \equiv y^2 \pmod{p}$, then $a \equiv \frac{ab}{b} \equiv \left(\frac{x}{y}\right)^2 \pmod{p}$, which would imply that b is a QR, which we assumed that it was not. Thus, $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = 1(-1) = -1 = \left(\frac{ab}{p}\right)$.

In the case that a and b are both QNR's, then we have to show that ab is a QR so that $\left(\frac{ab}{p}\right) = 1$. If we multiply a by all nonzero residues mod p , then we must get back all nonzero residues mod p , and we know that multiplying a by all the QR's give us all the QNR's, which constitute half the residues mod p . The other half of the residues mod p are QR's by Theorem 3.1. Thus, multiplying a by QNR's will give us QR's. In other words, if both a and b are QNR's, then $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = (-1)(-1) = 1 = \left(\frac{ab}{p}\right)$. $\square$

3.3 Primitive Roots

$\mathbb{Z}_p$ for primes p is special because it is a field, but it is also special because it always has a primitive root. To understand what a primitive root is, we first have to introduce the idea of an order of an number mod n .

Definition 3.1. If a and n are relatively prime integers, then the order of a mod n , denoted by $\text{ord}_n(a)$ is the smallest positive integer r such that $a^r \equiv 1 \pmod{n}$.

If $\text{ord}_n(a) = r$, then the result of taking consecutive powers of a mod n will only leave you with r numbers, because after the r th power, it will cycle back. Whenever there is an integer a such that a raised to consecutive powers will produce all residues relatively prime to n mod n , we call a a primitive root mod n . The reason it has to be relatively prime residues is that a is relatively prime to n itself, so any powers of it also must be relatively prime to n . We can phrase this condition more concisely if we use the Euler totient function.

Definition 3.2. The Euler totient function evaluated at a positive integer n , denoted by $\phi(n)$ is the number of positive integers less than or equal to n relatively prime to n .

Then, a is a primitive root mod n if $\text{ord}_n(a) = \phi(n)$. For a prime number p , this would imply that a is a primitive root of order $\text{ord}_p(a) = \phi(p) = p - 1$. It turns out that all prime numbers have primitive roots. To prove this, we first need two small lemmas.

Lemma 3.3.1. For relatively prime integers a and b , $\phi(ab) = \phi(a)\phi(b)$.

Proof. If a positive integer n is relatively prime to ab , then n is relatively prime to a and to b , and being relatively prime to a and to b are independent conditions since a and b are themselves relatively prime. Thus, $\phi(ab) = \phi(a)\phi(b)$. □

Lemma 3.3.2. For all positive integers n , we have:

$$\sum_{d|n} \phi(d) = n$$

Proof. We first consider the case where n has one prime factor, so it is of the form p^k and all of the divisors of p^k are of the form p^j . Therefore, we have:

$$\sum_{d|p^k} \phi(d) = \sum_{j=0}^k \phi(p^j)$$

It is easy to see that $\phi(p^j) = p^j - p^{j-1}$ for $j > 0$ because the only numbers relatively prime to p^j less than or equal to p^j are those that are not multiples of p and are in that range. Therefore, the sum becomes:

$$\sum_{j=0}^k \phi(p^j) = \sum_{j=0}^k p^j - p^{j-1} = 1 + p - 1 + p^2 - p + \cdots + p^k - p^{k-1} = p^k = n$$

Now suppose $n = p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}$. Then, all we have to do is multiply the sums $\sum_{d|p_i^{r_i}} \phi(d)$ for $i = 1, 2, \dots, k$, and we will obtain $\sum_{d|n} \phi(d)$. This is because the Euler totient function is multiplicative, product of these k sums is a unique product of k terms, each one from one of the sums $\sum_{d|p_i^{r_i}} \phi(d)$. Therefore,

$$\begin{aligned} \sum_{d|n} \phi(d) &= \left(\sum_{d|p_1^{r_1}} \phi(d) \right) \left(\sum_{d|p_2^{r_2}} \phi(d) \right) \cdots \left(\sum_{d|p_k^{r_k}} \phi(d) \right) \\ &= p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k} \\ &= n \end{aligned}$$

□

Theorem 3.1. For every prime p , there exists a primitive root mod p .

Proof. For every divisor d of $p-1$, let $\psi(d)$ be the number of residues mod p which have an order of d . On the one hand, every nonzero residue mod p has an order which is a divisor of $p-1$ because by Fermat's Little Theorem, every nonzero residue raised to the $p-1$ power is congruent to 1 mod p . Therefore, we can conclude that $\sum_{d|p-1} \psi(d) = p-1$.

Suppose for some $d | p-1$ that a has order d . Then, consider the polynomial $x^d - 1 \pmod{p}$, which by 2.2, has at most d solutions. However, $1, a, \dots, a^{d-1}$ are distinct solutions of that polynomial (we know they're distinct because d is the *order* of a). Obviously, any elements of order d must then be one of $1, a, \dots, a^{d-1}$, and those elements are the ones where a is raised to a power relatively prime

to d because otherwise, the order of that element would be less than d . Therefore, $\psi(d) = \phi(d)$ if there exists an element of order d at all.

What we have just shown is that if there exists an element of order d , there are $\phi(d)$ elements of order d . In other words, if $\psi(d) \neq 0$, $\psi(d) = \phi(d)$, so $\psi(d) = 0$ or $\psi(d) = \phi(d)$. This can be summarized as follows:

$$p-1 = \sum_{d|n} \psi(d) \leq \sum_{d|n} \phi(d) = p-1$$

with equality holding only when $\psi(d) = \phi(d)$ for all divisors of $p-1$. Since we can't have $p-1 < p-1$, we must have $\psi(d) = \phi(d)$ for all divisors of $p-1$. Thus, the number of primitive roots, which is $\psi(p-1)$ is equal to $\phi(p-1) > 0$. □

3.4 Euler's Criterion

Usually, we denote a primitive root mod p with the letter g . Thus, all the residues mod p are just $1, g, g^2, \dots, g^{p-1}$. If p is an odd prime, $p-1$ is even, which implies that the QR's are the even powers of g , and the QNR's are the odd powers of g . This provides another way to prove that the product of two QNR's is a QR as the product of two odd powers of g will obviously be an even power of g .

Euler's criterion uses primitive roots in a clever way to determine if a given number is QR mod p . It says the following:

Theorem 3.2. If a and p are relatively prime and p is prime, then

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$$

Proof. Let g be a primitive root and let $a \equiv g^r \pmod{p}$. If a is a QR, $\left(\frac{a}{p}\right) = 1$ and r is even, so $a^{\frac{p-1}{2}} \equiv (g^{p-1})^{\frac{r}{2}} \equiv 1 \pmod{p}$ since $\frac{r}{2}$ is an integer. On the other hand, if a is a QNR and r is odd, then $a^{\frac{p-1}{2}} \equiv (g^{r-1} \cdot g)^{\frac{p-1}{2}} \equiv 1 \cdot g^{\frac{p-1}{2}} \pmod{p}$. Since g is a primitive root, this is not congruent to 1. However, if we square $g^{\frac{p-1}{2}}$, we get 1, and the only other thing that squares to 1 is -1 , so $a^{\frac{p-1}{2}} \equiv g^{\frac{p-1}{2}} \equiv -1 \equiv \left(\frac{a}{p}\right) \pmod{p}$. □

If we wanted an algorithm to calculate the Legendre symbol with a computer, using Euler's criterion is a pretty good way to do that. Calculating exponents mod a given number are really fast to compute for a computer since we can repeatedly square the base and multiply those intermediate results together appropriately to obtain the particular power we want. This process closely resembles converting a number into binary. In particular, we are converting $\frac{p-1}{2}$ into binary. While this is a fast algorithm, we will see that using quadratic reciprocity will be even faster and more manageable to use to work out these problem by hand.

Finally, we can use primitive roots and Euler's criterion to figure out when there is a square root of -1 mod a prime p , or when $\left(\frac{-1}{p}\right) = 1$. By Euler's criterion, we know that $\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod{p}$. When $p-1$ is a multiple of 4, then the exponent of -1 is even, so $\left(\frac{-1}{p}\right) = 1$. When $p-1$ is not a multiple of 4, then the exponent of -1 will be odd, and $\left(\frac{-1}{p}\right) = -1$. In other words, this is equivalent to saying that -1 is QR when $p \equiv 1 \pmod{4}$ and QNR when $p \equiv 3 \pmod{4}$.

3.5 When is 2 a QR?

From the end of the last section, we saw that -1 is a QR mod a prime p if and only if p is 1 mod 4. There is a similar rule for when 2 is a QR. In particular, suppose that $s = \frac{p-1}{2}$, and consider the following equations:

$$\begin{aligned} 1 &= (-1)(-1)^1 \\ 2 &= 2(-1)^2 \\ 3 &= (-3)(-1)^3 \\ &\vdots \\ s &= (\pm s)(-1)^s \end{aligned}$$

Now we will multiply these equations and reduce mod p . On the left, we obviously get $s!$. On the right hand side, notice that $-1 \equiv 2\left(\frac{p-1}{2}\right) \equiv 2s \pmod{p}$. Similarly, $-3 \equiv 2(s-1) \pmod{p}$. Something similar occurs for odd numbers where they are congruent to 2 times s - minus something. On the other hand, $2 \equiv 2 \cdot 1 \pmod{p}$, $4 \equiv 2 \cdot 2 \pmod{p}$, and so on. The numbers $-1, 2, -3, \dots, \pm s$ are actually congruent to $2 \cdot 1, 2 \cdot 2, \dots, 2 \cdot s$ in some order (all mod p). Overall, we have:

$$\begin{aligned} s! &\equiv (2 \cdot 1)(2 \cdot 2) \cdots (2 \cdot s)(-1)^{1+2+\cdots+s} \\ s! &\equiv 2^s s! (-1)^{\frac{s(s+1)}{2}} \\ 1 &\equiv 2^s (-1)^{\frac{s(s+1)}{2}} \end{aligned}$$

Recall that $s = \frac{p-1}{2}$, so $2^s \equiv \left(\frac{2}{p}\right)$ by Euler's criterion. Furthermore, $\frac{s(s+1)}{2} = \frac{(p-1)(p+1)}{8}$. After a bit of rearranging, we thus get the identity:

$$\left(\frac{2}{p}\right) = (-1)^{\frac{(p-1)(p+1)}{8}} = (-1)^{\frac{p^2-1}{8}}$$

We can notice that when p is 1 or 7 (mod 8), the whole expression $\frac{p^2-1}{8}$ is even, but when p is 3 or 5 (mod 8), that expression is not even. In other words, 2 is a QR only when p is 1 or 7 mod 8.

4 Quadratic Reciprocity Law

4.1 Proof

Now, we will finally prove the Law of Quadratic Reciprocity, which says the following:

Theorem 4.1. Given distinct odd primes p and q , then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}$$

where $\left(\frac{p}{q}\right)$ and $\left(\frac{q}{p}\right)$ are Legendre symbols.

There are many different proofs of this theorem, but none of them are completely straightforward.

Proof. First, we will consider all the residues mod pq which are coprime to pq . There are exactly $\phi(pq) = (p-1)(q-1)$ of these. We can pair each residue with its negative. In particular, we can create $\frac{(p-1)(q-1)}{2}$ sets of the form $\{a, -a\}$ where a is a residue such that the union of these sets gives us all the residues mod pq coprime to pq .

We will now multiply one element from each set in 3 different ways. For each set, we have the choice of flipping a sign, so all 3 of these products will differ by at most a negative sign. Here are the three products:

- The first product will contain all coprime residues mod pq which are less than $\frac{p}{2}$ when reduced mod p . We can write this as:

$$A = \prod_{k=1}^{\frac{p-1}{2}} \left(\prod_{j=1}^{q-1} pj + k \right)$$

- The second product will contain all coprime residues mod pq which are less than $\frac{q}{2}$ when reduced mod q . We can write this as:

$$B = \prod_{k=1}^{\frac{q-1}{2}} \left(\prod_{j=1}^{p-1} qj + k \right)$$

- The third product will contain all coprime residues mod pq which are less than $\frac{pq}{2}$. We can write this as:

$$C = \prod_{\substack{a \bmod pq \\ \gcd(a, pq)=1 \\ a < \frac{pq}{2}}} a$$

We will calculate $A \bmod q$, $B \bmod p$, and $C \bmod$ both p and q , as well as the sign difference between A and $B \pmod{pq}$. Notice that in the innermost product for A , we have the product of $q-1$ nonzero residues which are separated by p each. Therefore, these numbers, in some order must just be equivalent to 1 to $q-1 \bmod q$, so the inner product is equivalent to $(q-1)! \bmod q$. Therefore, we can write:

$$\begin{aligned} A &\equiv \prod_{j=1}^{\frac{p-1}{2}} (q-1)! \\ &\equiv (q-1)!^{\frac{p-1}{2}} \pmod{q} \end{aligned}$$

Similarly, $B \equiv (p-1)!^{\frac{q-1}{2}} \pmod{p}$. Now, we can compare A and B . First, we will define a function $\lambda(r, s)$ where r is a residue mod p , s is a residue mod q , and $\lambda(r, s)$ gives us the residue mod pq which r and s determine (this is unique due to the Chinese Remainder Theorem). Then, A can be rewritten as:

$$\begin{aligned} A &= \prod_{r=1}^{\frac{p-1}{2}} \prod_{s=1}^{q-1} \lambda(r, s) \\ &= \left(\prod_{r=1}^{\frac{p-1}{2}} \prod_{s=1}^{\frac{q-1}{2}} \lambda(r, s) \right) \left(\prod_{r=1}^{\frac{p-1}{2}} \prod_{s=\frac{q+1}{2}}^{q-1} \lambda(r, s) \right) \end{aligned}$$

Similarly, we can write B as:

$$B = \left(\prod_{r=1}^{\frac{q-1}{2}} \prod_{s=1}^{\frac{p-1}{2}} \lambda(r, s) \right) \left(\prod_{r=1}^{\frac{q-1}{2}} \prod_{s=\frac{p+1}{2}}^{p-1} \lambda(r, s) \right)$$

Notice that the first factors for each of these expressions are equal; the order of the products is just flipped. For the second factor, we can use the fact that $\lambda(-r, -s) = -\lambda(r, s)$ and flip the indices, say for the second factor of A (let's call the second factor for A A_2 and the second factor B B_2). In A_2 , we are taking the product from $r = 1$ to $r = \frac{p-1}{2}$. If we instead take the product from $\frac{p+1}{2}$ to $p-1$, it has the effect of flipping the sign of $r \bmod p$, and we can do the same thing for s going from $\frac{q+1}{2}$ to $q-1$. In particular,

$$\begin{aligned} A_2 &\equiv \prod_{r=1}^{\frac{p-1}{2}} \prod_{s=\frac{q+1}{2}}^{q-1} \lambda(r, s) \\ &\equiv \prod_{r=\frac{p+1}{2}}^{p-1} \prod_{s=1}^{\frac{q-1}{2}} \lambda(-r, -s) \\ &\equiv \prod_{r=\frac{p+1}{2}}^{p-1} \prod_{s=1}^{\frac{q-1}{2}} -\lambda(r, s) \\ &\equiv (-1)^{\frac{(p-1)(q-1)}{4}} \cdot B_2 \end{aligned}$$

The exponent of -1 here comes from finding the total of number of factors multiplied out by the product. Finally, we have that $\frac{A}{B} = \frac{A_2}{B_2} = (-1)^{\frac{(p-1)(q-1)}{4}}$ or $A = (-1)^{\frac{(p-1)(q-1)}{4}} \cdot B$.

To calculate $C \bmod p$, notice that C is equivalent to the following:

$$C = \frac{(1 \cdot 2 \cdots (p-1)) \cdot ((p+1)(p+2) \cdots (2p-1)) \cdots \left((p(\frac{q-1}{2}) + 1)(p(\frac{q-1}{2}) + 2) \cdots (p(\frac{q-1}{2}) + \frac{p-1}{2}) \right)}{q \cdot 2q \cdot 3q \cdots \left(\frac{p-1}{2} \right) q}$$

In this fraction, the numerator contains all the residues mod pq less than $\frac{pq}{2}$ except for the multiples of p . However, we need to get rid of the multiples of q as well, and the denominator contains all the multiples of q in this range, canceling them out from the numerator. We can simplify this mod p , getting

$$\begin{aligned} C &\equiv \frac{(p-1)!^{\frac{q-1}{2}} \left(\frac{p-1}{2} \right)!}{q^{\frac{p-1}{2}} \left(\frac{p-1}{2} \right)!} \\ &\equiv \frac{(p-1)!^{\frac{q-1}{2}}}{q^{\frac{p-1}{2}}} \pmod{p} \end{aligned}$$

Similarly, we can reduce mod q to get:

$$C \equiv \frac{(q-1)!^{\frac{p-1}{2}}}{p^{\frac{q-1}{2}}} \pmod{q}$$

Now, comparing A and B with C , we get:

$$\begin{aligned}\frac{A}{C} &\equiv p^{\frac{q-1}{2}} \equiv \left(\frac{p}{q}\right) \pmod{q} \\ \frac{B}{C} &\equiv q^{\frac{p-1}{2}} \equiv \left(\frac{q}{p}\right) \pmod{p}\end{aligned}$$

Since A and C differ by at most a sign, $\frac{A}{C}$ is ± 1 when taken mod pq . Since $\left(\frac{p}{q}\right)$ is already ± 1 , this means that $\frac{A}{C} \equiv \left(\frac{p}{q}\right) \pmod{pq}$, and likewise, $\frac{B}{C} \equiv \left(\frac{q}{p}\right) \pmod{pq}$. Dividing these congruences, we get:

$$\frac{A}{B} \equiv \left(\frac{p}{q}\right) \left(\frac{q}{p}\right) \equiv (-1)^{\frac{(p-1)(q-1)}{4}} \pmod{pq}$$

Here, we used the fact that the Legendre symbols are ± 1 , so dividing them is the same as multiplying them. Now, we have Quadratic Reciprocity as a congruence mod pq , but we can turn it into an equation under real numbers because the congruence only involves powers of -1 mod pq , and powers of -1 behave exactly the same mod pq and in the real numbers, at least when multiplied. This proves the theorem. □

4.2 Application

Quadratic Reciprocity can be written in a more useful way, which provides more insight on how we can use it to calculate Legendre symbols efficiently. In particular:

- If either p or q is congruent to 1 mod 4, then $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$. This is because this would imply that the power of -1 in Theorem 4.1 is even, and so $\left(\frac{p}{q}\right) \left(\frac{q}{p}\right)$, from which it is clear that the two Legendre symbols are equal.
- If both p and q are congruent to 3 mod 4, then $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$. The proof of this case is basically the same as the last case, except the exponent of -1 ends up being odd.

With this in mind, we can calculate Legendre symbols with large numbers very fast. For example, consider $\left(\frac{1001}{99991}\right)$ (99991 is a prime). To use quadratic reciprocity, we need the numerator to be prime, so we can factor $1001 = 7 \cdot 11 \cdot 13$. Thus, we have:

$$\begin{aligned}\left(\frac{1001}{99991}\right) &= \left(\frac{7}{99991}\right) \left(\frac{11}{99991}\right) \left(\frac{13}{99991}\right) \\ &= \left(-\left(\frac{99991}{7}\right)\right) \left(-\left(\frac{99991}{11}\right)\right) \left(\left(\frac{99991}{13}\right)\right) \\ &= \left(\frac{3}{7}\right) \left(\frac{1}{11}\right) \left(\frac{8}{13}\right) \\ &= \left(-\left(\frac{7}{3}\right)\right) \left(\left(\frac{2}{13}\right)\right) \\ &= -\left(\frac{1}{3}\right) \left(\frac{2}{13}\right) \\ &= (-1) \cdot (-1) \\ &= 1\end{aligned}$$

This tells us that 1001 is in fact a QR mod 99991, which would be more tedious to calculate using Euler's criterion, at least by hand. All we do in each step of this process is for every Legendre symbol, we check if either the top or bottom is 1 mod 4, in which case we can flip it without any sign, and otherwise we need to add a negative sign. After that, we can reduce the top mod the bottom to get a smaller number. Of course, along the way, we also need to take care of factors of 2 and -1 , which aren't really applicable to Quadratic Reciprocity. We also need to make sure that the top of any Legendre symbol we use is a prime, so at the very least we need to factor the top of the Legendre symbol we are evaluating into primes.

We can also use Quadratic Reciprocity to obtain certain rules for when a fixed number (for now we will just consider a prime) is a QR mod an arbitrary prime, similar to how we obtained a rule for -1 and 2 . For example, we might be interested in when 5 is a QR. Since $5 \equiv 1 \pmod{4}$, we know that $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$, so 5 is a QR mod p when p is a QR mod 5 . In other words, p must be 1 or $4 \pmod{5}$.

As another example, we might want to know when 3 is a QR mod p . Because $3 \equiv 3 \pmod{4}$, we have two cases: when p is $1 \pmod{4}$ or $3 \pmod{4}$.

If $p \equiv 1 \pmod{4}$, we can flip $\left(\frac{3}{p}\right)$ to obtain $\left(\frac{p}{3}\right)$, which is 1 when p is $1 \pmod{3}$. Thus, if $p \equiv 1 \pmod{4}$ and $p \equiv 1 \pmod{3}$, which is equivalent to $p \equiv 1 \pmod{12}$, 3 is a QR mod p .

If $p \equiv 3 \pmod{4}$, $\left(\frac{3}{p}\right) = -\left(\frac{p}{3}\right)$, so we need $p \equiv 2 \pmod{3}$. Thus, if $p \equiv -1 \pmod{4}$ and $p \equiv -1 \pmod{3}$, which is equivalent to $p \equiv -1 \equiv 11 \pmod{12}$, we have that 3 is a QR mod p . Overall, 3 is a QR mod p if and only if p is 1 or $11 \pmod{12}$.

4.3 The Jacobi Symbol

To alleviate the problem of having to prime factorize the top of Legendre symbol before applying Quadratic Reciprocity, we can introduce the Jacobi Symbol.

Definition 4.1. For an integer a and a positive odd integer b , we define the Jacobi symbol of a over b , written $\left(\frac{a}{b}\right)$, where if $b = p_1 p_2 \cdots p_k$ where the p_i 's are primes (not necessarily distinct):

$$\left(\frac{a}{b}\right) = \left(\frac{a}{p_1 p_2 \cdots p_k}\right) = \left(\frac{a}{p_1}\right) \left(\frac{a}{p_2}\right) \cdots \left(\frac{a}{p_k}\right)$$

where the symbols on the right are Legendre symbols.

It is important to note that the Jacobi symbol does not tell us if a is a QR mod b . However, by definition, we know that if p is an odd prime, $\left(\frac{a}{p}\right)$ as a Jacobi symbol coincides in value with $\left(\frac{a}{p}\right)$ as a Legendre symbol. Therefore, $\left(\frac{a}{p}\right)$ as a Jacobi symbol is $+1$, then so is $\left(\frac{a}{p}\right)$ as a Legendre symbol, and this would tell us that a is a QR mod p . However, Jacobi symbols are easier to compute than Legendre symbols because we don't need to factorize anything, and Quadratic Reciprocity still holds.

Theorem 4.2. If a, b are coprime positive odd integers, then:

$$\left(\frac{a}{b}\right) \left(\frac{b}{a}\right) = (-1)^{\frac{(a-1)(b-1)}{4}}$$

Proof. Suppose $a = p_1 p_2 \cdots p_r$ for primes p_i (not necessarily distinct), and suppose $b = q_1 q_2 \cdots q_s$ for primes q_j (not necessarily distinct). Then, we have:

$$\begin{aligned}
\left(\frac{a}{b}\right) \left(\frac{b}{a}\right) &= \left(\frac{p_1 p_2 \cdots p_r}{q_1 q_2 \cdots q_s}\right) \left(\frac{q_1 q_2 \cdots q_s}{p_1 p_2 \cdots p_r}\right) \\
&= \prod_{i=1}^r \left(\frac{p_i}{q_1 q_2 \cdots q_s}\right) \left(\frac{q_1 q_2 \cdots q_s}{p_i}\right) \\
&= \prod_{i=1}^r \prod_{j=1}^s \left(\frac{p_i}{q_j}\right) \left(\frac{q_j}{p_i}\right) \\
&= \prod_{i=1}^r \prod_{j=1}^s (-1)^{\frac{(p_i-1)(q_j-1)}{4}} \\
&= (-1)^{\sum_{i=1}^r \sum_{j=1}^s \frac{(p_i-1)(q_j-1)}{4}}
\end{aligned}$$

Notice that

$$\begin{aligned}
\sum_{i=1}^r \sum_{j=1}^s \frac{(p_i-1)(q_j-1)}{4} &= \frac{1}{4} \left(\sum_{i=1}^r p_i - 1\right) \left(\sum_{j=1}^s q_j - 1\right) \\
&= \frac{1}{4} (p_1 + p_2 + \cdots + p_r - r)(q_1 + q_2 + \cdots + q_s - s)
\end{aligned}$$

We now claim the following:

$$\frac{p_1 + p_2 + \cdots + p_r - r}{2} \equiv \frac{a-1}{2} \pmod{2}$$

Since a is odd, $a \equiv 1, 3 \pmod{4}$. Suppose $a \equiv 1 \pmod{4} \implies \frac{a-1}{2} \equiv 0 \pmod{2}$. Furthermore, a must have an even number of prime factors which are $3 \pmod{4}$, so suppose that $2x$ of the p_i 's are $3 \pmod{4}$ while y of them are $1 \pmod{4}$. Then, we have

$$\begin{aligned}
p_1 + p_2 + \cdots + p_r - r &\equiv 3 \cdot 2x + 1 \cdot y - 2x - y \\
&\equiv 4x \\
&\equiv 0 \pmod{4}
\end{aligned}$$

This congruence implies that $\frac{p_1 + p_2 + \cdots + p_r - r}{2} \equiv 0 \equiv \frac{a-1}{2} \pmod{2}$.

The other case is when $a \equiv 3 \pmod{4} \implies \frac{a-1}{2} \equiv 1 \pmod{2}$. Furthermore, an odd number of the p_i 's must be $3 \pmod{4}$, say $2x+1$. Then, suppose there are y p_i 's which are $1 \pmod{4}$. Then, we have

$$\begin{aligned}
p_1 + p_2 + \cdots + p_r - r &\equiv 3(2x+1) + 1 \cdot y - 2x - 1 - y \\
&\equiv 4x + 2 \\
&\equiv 2 \pmod{4}
\end{aligned}$$

Dividing the congruence by 2 gives us $\frac{p_1 + p_2 + \cdots + p_r - r}{2} \equiv 1 \equiv \frac{a-1}{2} \pmod{2}$.

In a similar fashion, we have that $\frac{q_1+q_2+\dots+q_s}{2} \equiv \frac{b-1}{2} \pmod{2}$. Combining these two facts together, we get

$$\left(\frac{p_1+p_2+\dots+p_r-r}{2}\right)\left(\frac{q_1+q_2+\dots+q_s-s}{2}\right) \equiv \left(\frac{a-1}{2}\right)\left(\frac{b-1}{2}\right) \pmod{2}$$

Finally, this implies that

$$\begin{aligned} \left(\frac{a}{b}\right)\left(\frac{b}{a}\right) &= (-1)^{\sum_{i=1}^r \sum_{j=1}^s \frac{(p_i-1)(q_j-1)}{4}} \\ &= (-1)^{\left(\frac{p_1+p_2+\dots+p_r-r}{2}\right)\left(\frac{q_1+q_2+\dots+q_s-s}{2}\right)} \\ &= (-1)^{\left(\frac{a-1}{2}\right)\left(\frac{b-1}{2}\right)} \\ &= (-1)^{\frac{(a-1)(b-1)}{4}} \end{aligned}$$

This proves the identity. □

Now we can try evaluating that Legendre symbol from the last section without having to factorize anything. In particular, we have

$$\begin{aligned} \left(\frac{1001}{99991}\right) &= \left(\frac{99991}{1001}\right) = \left(\frac{892}{1001}\right) = \left(\frac{4}{1001}\right)\left(\frac{223}{1001}\right) = \left(\frac{1001}{223}\right) = \left(\frac{109}{223}\right) = \left(\frac{223}{109}\right) = \left(\frac{114}{109}\right) \\ &= \left(\frac{2}{109}\right)\left(\frac{57}{109}\right) = (-1)\left(\frac{109}{57}\right) = -\left(\frac{52}{57}\right) = -\left(\frac{13}{57}\right) = -\left(\frac{57}{13}\right) = -\left(\frac{5}{13}\right) = -\left(\frac{13}{5}\right) \\ &= -\left(\frac{3}{5}\right) = -\left(\frac{5}{3}\right) = -\left(\frac{2}{3}\right) = 1 \end{aligned}$$

In this case, we have more steps than when we just used the standard Legendre symbol method, but since we don't have to prime factor anything, this actually works better for larger numbers.

5 Sums of Squares

The main question we will be addressing in this section is when a number can be written as the sum of multiple of perfect squares, so naturally, we would first like to know which numbers can be written as the sum of exactly 2 perfect squares. To gain some intuition for this, it is useful to consider the Gaussian integers.

5.1 Gaussian Integers

The Gaussian Integers, denoted $\mathbb{Z}[i]$, are the set of complex numbers $a + bi$, where $a, b \in \mathbb{Z}$, which can also be written as:

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$$

With Gaussian integers, we can interpret the sum of two squares easily. In particular, for integers x and y , we have that $x^2 + y^2 = (x + yi)(x - yi)$, which is the product of two complex conjugates. This leads to the following proposition:

Proposition 5.1. If a and b are positive integers which can both be written as the sum of 2 squares, then so can ab .

Proof. Suppose that

$$\begin{aligned}a &= x^2 + y^2 = (x + yi)(x - yi) \\ b &= z^2 + w^2 = (z + wi)(z - wi)\end{aligned}$$

Multiplying a and b gives us the following:

$$\begin{aligned}ab &= (x + yi)(z + wi)(x - yi)(z - wi) \\ &= (xz - yw + i(xw + yz))(xz - yw - i(xw + yz)) \\ &= (xz - yw)^2 + (xw + yz)^2\end{aligned}$$

which is a sum of two integer squares. □

Although this theorem can be stated purely algebraically without Gaussian integers, it provides some more insight into their structure. In particular, we used the fact that the product of the conjugates of two complex numbers is conjugate of their products, which is why $(x - yi)(z - wi)$ turns out to be the conjugate of $(x + yi)(z + wi)$.

5.2 Some Terminology

In many ways, $\mathbb{Z}[i]$ can be treated like the integers. In particular, it is known as a Euclidean Domain, which also implies that it is a Unique Factorization Domain. We won't worry about these terms too much, but we will discuss some of the properties they imply for $\mathbb{Z}[i]$, but we won't go deep into their proofs, because it is more algebraic than number theoretic.

First of all, the norm of a Gaussian integer x , denoted $\|x\|$, is the square of its absolute value, or $|x|^2$. Also, if a Gaussian integer has a norm of 1, we call it a unit. The only units in the Gaussian integers are ± 1 and $\pm i$.

Furthermore, while we won't prove this, the Gaussian integers have unique factorization. In particular, if $a = x_1x_2 \dots x_n = y_1y_2 \dots y_m$ where the x_i 's and y_i 's are Gaussian integers which are not units, then each $m = n$, and each x_i is equal to some y_j times some unit. In other words, factorization is unique up to multiplying by units. For example, $5 = (2 + i)(2 - i) = (i(2 + i))(-i(2 - i)) = (-1 + 2i)(-1 - 2i)$, but these are not different factorizations because each factor in the different factorizations are just off by units (i and $-i$).

5.3 Fermat's Two Square Theorem

Now we are ready to prove the famous Two Square Theorem, which is often attributed to Fermat. It is also often called Fermat's Christmas Theorem.

Theorem 5.1. If p is a prime which is congruent to 1 mod 4, then it can be written as the sum of two perfect squares.

Proof. Since $p \equiv 1 \pmod{4}$, $\left(\frac{-1}{p}\right) = 1$, so there exists an integer x such that $x^2 \equiv -1 \pmod{p} \implies x^2 + 1 = kp$ for some integer k . $x^2 + 1$ can be factored over the Gaussian integers into primes, say $q_1, q_1, \dots, q_r$, giving

$$q_1 q_2 \cdots q_r = kp$$

Remember that the q_i 's are primes over $\mathbb{Z}[i]$, not over $\mathbb{Z}$. Furthermore, none of the q_i 's can be a pure real or imaginary number because this would scale the product of the q_i 's by some real number and make the imaginary part of their product greater than 1, which it isn't. Therefore, we can choose one of the q_i 's such that it divides p . Suppose that $q_i = z + wi$.

In other words, we have that $z + wi \mid p$, so $\|z + wi\| \mid \|p\| \implies z^2 + w^2 \mid p^2$. Since $z + wi$ is not a unit, we also have that $z^2 + w^2 > 1$. Furthermore, since $z + wi$ is a prime factor of p not equal to p , $z^2 + w^2 < p^2$. The only divisor of p between 1 and p^2 is p itself, so $z^2 + w^2 = p$. Thus, every prime which is 1 mod 4 can be written as the sum of 2 squares. □

We can also prove that every such prime can be written in exactly one way as the sum of two squares, up to signs and ordering.

Theorem 5.2. If p is a prime such that $p = a^2 + b^2 = c^2 + d^2$ such that $0 < a \leq b$, $0 < c \leq d$, then $a = c$ and $b = d$.

Proof. p can be factored in $\mathbb{Z}[i]$ as $(a + bi)(a - bi)$. Since $\|a + bi\| = \|a - bi\| = p$, we must have that $a \pm bi$ are primes in $\mathbb{Z}[i]$ (since if it could factor further, one of the factors would have a norm of 1), so $(a + bi)(a - bi)$ is the prime factorization of p in $\mathbb{Z}[i]$. Likewise, $(c + di)(c - di)$ is also the prime factorization of p . Therefore, these prime factorizations must only differ by units, and because of the ordering we've imposed on a, b, c, d , we must have that $a + bi = c + di$ and $a - bi = c - di$. In other words, we don't have to worry about anything differing by units. Thus, we have that $a = c$ and $b = d$. □

5.4 Examples

Here are just a couple of examples of primes which are 1 mod 4 and their representations as a sum of two squares.

$$\begin{aligned} 5 &= 2^2 + 1^2 \\ 13 &= 3^2 + 2^2 \\ 17 &= 4^2 + 1^2 \\ 29 &= 5^2 + 4^2 \\ 37 &= 6^2 + 1^2 \\ 41 &= 5^2 + 4^2 \\ 53 &= 7^2 + 2^2 \end{aligned}$$

In the more general study of representing two numbers, it is useful to note that $2 = 1^2 + 1^2$ and also that no prime which is 3 mod 4 can be written as the sum of two squares. This is simply because any square is either congruent to 0 or 1 mod 4, so adding 2 squares will give you something 0, 1, or 2 mod 4, but never 3.

5.5 The General Case

We will now classify all positive integers which can be written as the sum of two squares.

Theorem 5.3. A positive integer n can be written as a sum of two squares if and only if all primes $3 \bmod 4$ in its factorization appear with an even exponent.

Proof. Suppose $n = 2^k \cdot p_1^{a_1} \cdots p_r^{a_r} \cdot q_1^{b_1} \cdots q_s^{b_s}$, where the p_i 's are distinct primes $1 \bmod 4$, and the q_j 's are distinct primes $3 \bmod 4$.

To prove the backward direction, suppose that all the b_j 's are even. All the primes $2, p_1, \dots, p_r$ can be written as the sum of 2 squares, so by Proposition 5.1, $2^k p_1^{a_1} \cdots p_r^{a_r}$ can as well. Let's say it is equal to $X^2 + Y^2$. Furthermore, $q_1^{b_1} \cdots q_s^{b_s}$ is a perfect square, say Z^2 . Thus, $(XZ)^2 + (YZ)^2 = n$.

To prove the forward direction, suppose that one of the b_j 's are odd. Without loss of generality, we can take b_1 to be odd. Suppose that $x^2 + y^2 = n$. Reducing mod q_i , we get

$$\begin{aligned} x^2 + y^2 &\equiv 0 \pmod{q_i} \\ x^2 &\equiv -y^2 \pmod{q_i} \end{aligned}$$

If it is not the case that $x \equiv y \equiv 0 \pmod{q_i}$, then we have that $-y^2$ is a QR mod q_i , so $\left(\frac{-y^2}{p}\right) = \left(\frac{-1}{p}\right) = 1$, but this is not true because $p \equiv 3 \pmod{4}$. Thus, $x = x_1 q$ and $y = y_1 q$ for integers x_1, y_1 . This gives that $x_1^2 + y_1^2 = \frac{n}{q_1^2}$. $\frac{n}{q_1^2}$ still has an odd exponent of q_1 , so we can continue this process until we reach a number n' which has an exponent of 1 on q_1 , at which point we will have integers x' and y' such that $x'^2 + y'^2 = n'$. If we try this process again, we will see that $q_1 \mid x'$ and $q_1 \mid y'$, so $q_1^2 \mid x'^2 + y'^2 = n'$, but this is not true because n' only has an exponent of 1 on q_1 . Thus, we have contradicted the existence of a representation of n as the sum of two squares, proving the theorem. □

5.6 Some Related Results

Here are some results related to the Two Square Theorem, but which we will not prove here. First of all, we might want to know what integers can be written as the sum of 3 or 4 squares. In fact, we have the following:

Theorem 5.4. An integer n can be written as the sum of 3 perfect squares if and only if it is not of the form $4^m(8k+7)$ for integers m and k .

Theorem 5.5. All positive integers can be written as the sum of 4 perfect squares.

A similar result to Theorem 5.6 exists for triangular numbers, where the n th triangular number is $\frac{n(n+1)}{2}$.

Theorem 5.6. All positive integers can be written as the sum of 3 perfect squares.